

Maximum likelihood estimation

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Recap

Definition: The *likelihood* $L(Model) = P(Data|Model)$ of a model is the probability of the observed data, given that we assume a certain model and certain values for the parameters that define that model.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Model: $Y_i \sim \text{Bernoulli}(\pi)$, and $\hat{\pi} = 0.9$
- + Data: $y_1, \dots, y_5 = T, T, T, T, H$
- + Likelihood: $L(0.9) = P(y_1, \dots, y_5 | \pi = 0.9) = 0.00009$

$$= (1 - 0.9)^4 (0.9)$$

For any value of π : $(1 - \pi)^4 \pi$
for T, T, T, T, H

Recap

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\pi) = (1 - \pi)^4(\pi)$
- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

Recap

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

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How do I maximize functions?

Take derivative, set $\underline{\quad} = 0$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(\text{Heads})$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\pi) = (1 - \pi)^4(\pi)$
- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

Step 1: Write down the likelihood

$$L(\pi) = (1 - \pi)^4(\pi)$$

$$\log(ab) = \log(a) + \log(b)$$

$$\log(a^b) = b\log(a)$$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

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- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

Step 2: Take the log

$$\begin{aligned} \log L(\pi) &= \log((1 - \pi)^4 \pi) \\ &= \log((1 - \pi)^4) + \log(\pi) \\ &= 4 \log(1 - \pi) + \log(\pi) \end{aligned}$$

$\log(\dots)$ is an increasing function

so if $L(n)$ is big, $\log(L(n))$ is
also big

(vice-versa)

$\Rightarrow$ value of n that maximizes $\log L(n)$

$\equiv$
value of n that maximizes $L(n)$

Maximum likelihood estimation with calculus

Maximum likelihood estimation: pick the parameter estimate that maximizes the likelihood.

Coin example: flip a coin 5 times, with $\pi = P(Heads)$

- + Observed data: T, T, T, T, H
- + Likelihood: $L(\pi) = (1 - \pi)^4(\pi)$
- + Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

Step 3: maximize

$$\log L(\pi) = 4 \log(1 - \pi) + \log(\pi)$$

How do we find where maxima/minima occur for a function?

take derivative, set $= 0$

Maximum likelihood estimation with calculus

+ Observed data: T, T, T, T, H

+ Likelihood: $L(\pi) = (1 - \pi)^4(\pi)$

+ Choose $\hat{\pi}$ as value of π that maximizes $L(\pi)$

$$\frac{1}{S} = \frac{\# H}{\# \text{ flips}}$$

$$\boxed{\hat{\pi} = \frac{1}{S}}$$

Maximum
likelihood
estimate
(MLE)

Step 3: maximize -- take first derivative and set equal to 0

$$\log L(\pi) = 4 \log(1 - \pi) + \log(\pi)$$

$$\frac{d}{d\pi} \log L(\pi) = \frac{4}{1-\pi} (-1) + \frac{1}{\pi}$$

$$= -\frac{4}{1-\pi} + \frac{1}{\pi} \stackrel{\text{set}}{=} 0$$

$$\Rightarrow \frac{1}{\pi} = \frac{4}{1-\pi}$$

$$\Rightarrow \frac{1-\pi}{\pi} = 4$$

$$\Rightarrow \frac{1}{\pi} - 1 = 4 \Rightarrow \frac{1}{\pi} = 5 \Rightarrow \pi = \frac{1}{5}$$

So $L(\pi)$ is maximized

at $\pi = \frac{1}{5}$

Class activity

Work on the activity (handout) with a neighbor, then we will discuss as a class

Class activity

$$+ P(Y_i = 0) = \pi_0$$

$$+ P(Y_i = -1) = 2\pi_0$$

$$+ P(Y_i = 1) = 1 - 3\pi_0$$

Observe data $y_1, y_2, y_3, y_4, y_5, y_6$
 $-1, -1, 0, 1, 0, -1$.

$$L(\pi_0) = ? \quad (2\pi_0)^3 \quad (\pi_0)^2 \quad (1-3\pi_0)$$

(3 -1's) (2 0's) (1 1)

$$L(\pi_0) = P(y_1, \dots, y_6 | \pi_0) \stackrel{\text{(independence)}}{=} P(y_1 | \pi_0) P(y_2 | \pi_0) \dots P(y_6 | \pi_0)$$

$$= P(-1 | \pi_0) P(-1 | \pi_0) P(0 | \pi_0) P(1 | \pi_0) P(0 | \pi_0) P(-1 | \pi_0)$$

$$= P(-1 | \pi_0)^3 P(0 | \pi_0)^2 P(1 | \pi_0)$$

$$= (2\pi_0)^3 \pi_0^2 (1-3\pi_0) = 8\pi_0^5 (1-3\pi_0)$$

$$\log L(\pi_0) = \log(8) + 5\log(\pi_0) + \log(1-3\pi_0)$$

Class activity

$$\log L(\pi_0) = \log(8) + 5 \log(\pi_0) + \log(1 - 3\pi_0)$$

$$\frac{d}{d\pi_0} \log L(\pi_0) = \frac{5}{\pi_0} + \frac{1}{1-3\pi_0} (-3) \stackrel{\text{set}}{=} 0$$

$$\Rightarrow \frac{5}{\pi_0} = \frac{3}{1-3\pi_0} \Rightarrow \frac{1-3\pi_0}{\pi_0} = \frac{3}{5}$$

$$\Rightarrow \frac{1}{\pi_0} - 3 = \frac{3}{5} \Rightarrow \frac{1}{\pi_0} = \frac{3}{5} + 3 = \frac{18}{5}$$

$$\Rightarrow \pi_0 = \frac{5}{18}$$

$$\text{So, } \pi_0 = \frac{5}{18} \text{ maximizes } L(\pi_0)$$

$$\boxed{\hat{\pi}_0 = \frac{5}{18}}$$

(MLE)

Maximum likelihood in more generality

$$Y_i = 0 \text{ or } 1$$

$$\pi = P(Y_i = 1)$$

Suppose that $Y_i \sim \text{Bernoulli}(\pi)$. We observe n independent observations $Y_1, \dots, Y_n$ and want to estimate π .

Step 1: Write down the likelihood

✚ Let k be the number of times $Y_i = 1$ in the data

$$L(\pi) =$$

$$P(Y_i = y) = \begin{cases} 1 - \pi & y = 0 \\ \pi & y = 1 \end{cases}$$

$$P(Y_i = 0) = 1 - \pi \quad P(Y_i = 1) = \pi$$

$$= \pi^y (1 - \pi)^{1-y}$$

$$y = 0 \rightarrow \pi^0 (1 - \pi)^{1-0} = 1 - \pi$$

$$y = 1 \rightarrow \pi^1 (1 - \pi)^{1-1} = \pi$$

$$P(Y_1, Y_2, Y_3, \dots, Y_n | \pi) = P(Y_1 | \pi) P(Y_2 | \pi) \dots P(Y_n | \pi)$$

$$= \prod_{i=1}^n P(Y_i | \pi) \leftarrow \text{product of } n \text{ probabilities}$$

$$= \prod_{i=1}^n \pi^{Y_i} (1 - \pi)^{1-Y_i}$$

Maximum likelihood in more generality

Suppose that $Y_i \sim \text{Bernoulli}(\pi)$. We observe n independent observations $Y_1, \dots, Y_n$ and want to estimate π .

Step 1: Write down the likelihood

✚ Let k be the number of times $Y_i = 1$ in the data $= \sum_{i=1}^n Y_i$

$$\begin{aligned}
 L(\pi) &= \prod_{i=1}^n \pi^{Y_i} (1-\pi)^{1-Y_i} \\
 &= \pi^{Y_1 + Y_2 + \dots + Y_n} (1-\pi)^{(1-Y_1) + (1-Y_2) + \dots + (1-Y_n)} \\
 &= \pi^{\sum_{i=1}^n Y_i} (1-\pi)^{\sum_{i=1}^n (1-Y_i)} = \pi^{\sum_{i=1}^n Y_i} (1-\pi)^{n - \sum_{i=1}^n Y_i} \\
 &= \pi^k (1-\pi)^{n-k} \\
 L(\pi) &= \pi^k (1-\pi)^{n-k}
 \end{aligned}$$

Maximum likelihood in more generality

Step 1: Write down the likelihood

$$L(\pi) = \pi^k (1 - \pi)^{n-k}$$

Step 2: Take the log

$$\log L(\pi) = k \log(\pi) + (n-k) \log(1-\pi)$$

$$\frac{d}{d\pi} \log L(\pi) = \frac{k}{\pi} + \frac{n-k}{1-\pi} (-1) \stackrel{\text{set}}{=} 0$$

$$\hat{\pi} = \frac{k}{n}$$

$$\boxed{\hat{\pi} = \frac{k}{n}}$$

($k = \# \{i=1\}$, $n = \text{total \# of observations}$)

e.g. $\frac{\# H}{\# \text{ flips}}$

Maximum likelihood in more generality

So our maximum likelihood estimate is $\hat{\pi} = \frac{k}{n}$, the sample proportion

+ Example data: T, T, T, T, H

+ This implies that $\hat{\pi} = \frac{1}{5} = 0.2$